



Extending Information Delivery Specifications for digital building permit requirements

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ABSTRACT

This paper examines the potential for extending the Information Delivery Specification (IDS) schema to facilitate its integration into the building permit process. IDS is an open specification for defining and verifying information requirements for digital building models. Building authorities have already begun utilising IDS to define information requirements for automated code compliance checks. However, this requires the filtering of elements beyond the current scope of IDS. Such filtering functionality also has the potential to facilitate simple code compliance checks. This paper presents concepts for extending IDS to define information requirements for escape route analysis and code compliance checks for Austrian fire resistance regulations. The main extension was the filtering of elements depending on characteristics of related elements. Nesting existing functionalities enabled the schema to remain simple. A prototype implementation demonstrated that minor extensions to the IDS schema can significantly enhance its functionality, paving the way for broader applications of IDS.

1. Introduction

The use of Building Information Modelling (BIM) to verify the quality and accuracy of data continues to grow. In addition to the initial use of BIM for quality assurance during the design phase, building authorities are now implementing semi-automated compliance checks based on BIM models (Amor and Dimyadi, 2021; Beach et al., 2020; Noardo et al., 2022a; Kim et al., 2020; Shahi et al., 2019; Ullah et al., 2020). This automation step is expected to yield benefits in terms of improved design quality and reduced time for the building permit process (Eastman et al., 2009). A fundamental requirement for the functioning of automated model-based checking systems is the availability of the required information in BIM models (Amor and Dimyadi, 2021; Eastman et al., 2009; Noardo et al., 2022b). How to define this information (van Berlo et al., 2019; Tomczak et al., 2022; Klusmann et al., 2020; Cavka et al., 2017) and how to process it in the BIM workflow (Klusmann et al., 2020; Mellenthin Filardo et al., 2023b,a) are topics that have already been discussed in research. Tomczak et al. (2022) compared different options and formats for defining information requirements and found that in 2022 the most commonly used form was text-based documents, such as PDF. This is often due to their role as contractual documents. Spreadsheets are often used as a slightly more structured form of information requirements for exchange between project participants. However, the main shortcoming of the two established methods is the lack of automated processing capabilities.

Users have to manually recreate the information in BIM authoring or checking software. Recently, buildingSMART International introduced the Information Delivery Specification (IDS) (buildingSMART International, 2024b), an open standard that aims to bridge the gap between the definition and verification of information requirements. IDS is a computer-interpretable schema for semantic alphanumeric information requirements. It is designed to represent information for BIM models in the ISO-standardised IFC (Industry Foundation Classes) schema (International Organization for Standardization, 2024), also published by buildingSMART International. IDS covers two cases in the BIM workflow: defining information requirements and checking IFC models against those requirements, eliminating the manual step of recreating the information requirements in the software application used. Tomczak et al. (2022) identified IDS as the most beneficial method for checking information requirements.

IDS is gradually establishing itself in BIM processes and has already been implemented by several software applications (e.g. Plannerly, BIMQ, Solibri Office and BIMcollab Zoom). In research, IDS is being considered in various use cases such as providing general exchange information requirements to all project stakeholders (Austrian research project AIA4ALL (Austrian Forschungsförderungsgesellschaft FFG, 2024)), the specification of circularity data (Tomczak et al., 2024) and structural health monitoring data (Gragnaniello et al., 2024), and

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the digital building permission (de Marco et al., 2024; Ciccone et al., 2023). The building authority of Vienna is already adapting to IDS for providing their information requirements (buildingSMART Austria, 2024), as open formats are a basic requirement for them. However, they do not fully exploit the possibilities of IDS. In current practice, it is common to define information requirements for all elements of a given entity, class or type (e.g. all IfcDoor elements must contain certain properties). In many cases not all elements of a particular entity require this information. Take the example of the escape route analysis (Fischer et al., 2023) developed for the openBIM permit process of the Vienna Building Authority. Some properties are required for doors along escape routes and others for doors at the start of escape routes (flat entrance doors). It is not necessary to add this information to all doors. In contrast, when using classical entity-based information requirements, each modelled door requires all specified properties of the entity, otherwise checking tools will indicate errors. IDS has functionalities to define information requirements more precisely. Applicable elements can be selected by their characteristics such as attributes, properties, materials, and relations. However, whether elements require information or not is often not explicitly given by their own characteristics, but can be determined by characteristics of related elements. For the escape route analysis example, flat entrance doors need to be identified. Instead of introducing an additional property, flat entrance doors are identified by having a space of a flat on one side and an arbitrary space on the other side. While IDS provides the functionality to specify properties of elements and also to request related elements, it is not possible to specify properties of related elements. Since both required functionalities are already individually contained in IDS, an extension of the schema to combine these functionalities seems promising.

Such functionality can be used not only to check compliance with information requirements, but also has the potential to represent actual code compliance checks. Representing code compliance checks in IDS would have the advantage for building authorities of reusing an open format already in use for another purpose. The reduction in the number of systems or formats required for digital building permitting simplifies the process and facilitates its management, as fewer different competencies are necessary. The competencies developed for one purpose can be applied to another. This is crucial for the implementation of the digital building permit process, for which numerous new competencies are required in building authorities (Ullah et al., 2020). In addition, the availability of an open source rule language is often called for in the discussion of how to represent building regulations in a computer-interpretable form (Stepien et al., 2023; Ghannad et al., 2019). Nuyts et al. (2024) have provided an analysis of the use of standard IDS for general code compliance checking. They observed that while information availability, target values, and relational constraints can be represented generally, mathematical and conditional clauses remain unaddressed. Our proposed extension seeks to enhance the capabilities of IDS in handling relational constraints by enabling the definition of characteristics pertaining to related elements.

Research on extending IDS is rare as it is a very new specification. However, Kremer and Beetz (2023) have already identified potential extensions to its checking capabilities. They presented a possible extension to include geometric and topological checking in addition to semantic checking. Kremer and Beetz (2023) achieved this by linking external validation services in the IDS to perform the required calculations. While the concept of offloading extensions from the IDS schema to external sources provides flexibility and independence of the extensions, the checking process becomes less transparent. Furthermore, since these extensions are not part of the standardised schema, users are always dependent on the availability of the corresponding checking algorithms. While the described geometric and topological checking functionalities are explicitly excluded from the standard IDS functionality, the use cases we considered concern functionalities very similar to the current scope of IDS. Therefore, it seems promising to include them directly in the IDS specification.

In conclusion, two core research questions emerged from the introduction of IDS in the building permit process:

- Can IDS be modified to define specifications based on characteristics of related elements?
- What potential does IDS have for code compliance checking?

In this study, we use two related use cases to answer the specified research questions. The already mentioned complex checking rule for escape route analysis of the Vienna Building Authority provides specific information requirements depending on the related elements. In addition, Austrian building regulations concerning the fire resistance of elements are considered. These regulations were selected because they define alphanumeric information requirements and include dependencies on other elements. More complex regulations concerning the geometry or topology of elements are not considered, as IDS is explicitly designed for alphanumeric information requirements. The following Section 2 describes the required materials and the steps to extend IDS based on Loibl (2024). Section 3 presents the results of applying the extension to test models. These results are discussed in Section 4, which highlights the capabilities and limitations of the proposed extensions and describes how they point the way forward for IDS-based code compliance checking in general. Finally, Section 5 concludes with the implications of our work and possible future developments.

2. Material and methods

The proposed IDS extension is intended to be implemented in digital building permit processes. The core feature of a digital building permit process is the replacement of 2D application plans with digital data and the use of digital tools to automate parts of the process (Noardo et al., 2022a). In this research, the openBIM permit process of the Vienna Building Authority is used as a basis, which has replaced 2D application plans with BIM models. It was developed during the EU-funded research project BRISE-Vienna (UIA – Urban Innovative Actions, 2019). The development included the digital transformation of the existing processes (Urban et al., 2024), including identifying the required data, creating automated code compliance checks (Fischer et al., 2023), and developing concepts to involve non-technical stakeholders into the digital process using modern technologies such as augmented reality (Gerger et al., 2023; Schranz et al., 2021). Automated compliance checking requires two digital data sources: a computer-readable representation of the building code, and a BIM model containing the application information. The building code requirements represent the real requirements that the actual building must meet. From these real requirements, information requirements are derived for the BIM model that are needed to check the BIM model against the real requirements, e.g. the occupancy number is needed to check the person-dependent width of escape routes. The basis for all model-based steps was therefore a detailed definition of the required data. This information is contained in the Building Application Model (BAM). This is not a separate domain model. Rather, the architecture domain model is extended by the information requirements of the digital building permit use case according to the European standard EN 17412-1 (Austrian Standards International, 2021). The Building Application Model is submitted to the building authority as an IFC file to apply for a building permit. IFC is well established in the construction industry and has been used in many digital building permit initiatives. Therefore, this research investigates the capability and required extensions of IDS to define information requirements for the Building Application Model in IFC and to check it against the building code.

IDS is a standard from buildingSMART International with the purpose of being an open and lightweight schema for computer-interpretable information requirements (buildingSMART International, 2024b). The first official version of IDS (1.0) was published in June 2024. This official version was used as the basis for this research. The IDS schema is XML-based and defined by an XML Schema Definition (XSD). IDS is structured as a list of specifications that define the information requirements for specific elements of an IFC model. Such specifications are divided into two parts: applicability and requirements. The

applicability part defines the elements affected by the specification. The requirements part defines the actual information requirements for the applicable elements. IDS defines the conditions for both the applicability and requirements parts by facets. Six different facets are available, which are closely linked to the IFC schema. The facets can deal with the IFC entity, attributes, properties, external classifications, materials and relations of model elements. By combining the facets, very specific information requirements can be formulated for very specific elements. Of particular interest for the proposed IDS extension is the PartOf facet, which implements relations to other model elements or entities. It specifies the related IFC entity (mandatory) and the relation referencing the entity (optional), e.g. a wall must be related to an IfcBuildingStorey via the IfcRelContainedInSpatialStructure relation. Consequently, the PartOf facet is the basis for defining information requirements depending on related elements.

The first step towards an IDS extension was to analyse the considered use cases to identify the challenges and requirements for IDS. Concepts for extending the IDS schema were then developed and incorporated into the IDS schema definition. The extensions were validated by applying the extended IDS files to test models using a developed prototype system. Finally, based on the results of the two specific use cases, extensions for the general use of IDS for code compliance checking are conceived.

2.1. Escape route analysis information requirements

In this research, we identified potential IDS extensions on the example of the complex regulation for escape route analysis in the Austrian building code. In Austria, the nationally applicable OIB guidelines (Austrian Institute of Construction Engineering, 2023c) provide the definition and requirements of escape routes. These guidelines consist of several parts. The OIB 2 guideline (Austrian Institute of Construction Engineering, 2023a) for general fire protection specifies, for example, the necessity of escape routes, start and end locations, safe spaces, and the maximum route length. Additionally, the OIB 4 guideline (Austrian Institute of Construction Engineering, 2023b) defines geometric requirements such as minimum width and height. These requirements served as the foundation for a semi-automated checking rule for escape route analysis developed in BRISE-Vienna (Fischer et al., 2023). The checking rule implemented the concept of two separate automated steps of generating escape routes from implicitly contained information in BIM models and validating a selected escape route network (Fischer et al., 2024a). A manual step was necessary between these steps to comply with the legal system in Vienna. Escape route generation provides a pool of legally compliant escape routes. In a subsequent manual step, designers can select the desired network of escape routes for the entire building from this collection. This selection is realised by defining the sequence of doors for each escape route. This information is stored in the Building Application Model by a reference from the current door to the next door on the escape route. Subsequently, escape route validation evaluates the entire network, considering the interaction between individual routes, such as the common flow of people, which is relevant for the width requirements of doors, corridors, and stairs.

The foundation of the escape route analysis is the identification of specific categories of spaces and doors, which require different information. The objective of the information requirements for the escape route analysis within BRISE-Vienna was to utilise properties that were also required for other regulations to minimise the overall number of specifications. The escape route analysis must identify routes from each space that allows occupants to reach a safe evacuation point outside the building. An exception is made for flats that extend over a maximum of two levels with no direct exit to the outside. In such cases, no individual escape route for spaces inside flats are required. Rather, the escape route begins at the flat entrance doors. Consequently, the spaces of flats must be excluded from the escape route generation process, and the

flat entrance doors must be identified as the start point for the escape routes. It is also necessary to pay special attention to safe staircases. These are interim destinations of the escape route analysis, as the legal length of an escape route is measured from its start point until reaching a safe staircase, if one exists. The route through the safe staircase is not part of that. Finally, parking spots are the third special space for the escape route analysis. In case of emergency, it has to be assumed that all parking spots are occupied, therefore they are obstacles for the escape route analysis.

Table 1 summarises the information requirements for escape route analysis, defining the applicable elements in the left column and the requirements (property set and property or predefined type) in the right column. The summary includes a categorisation of whether the requirement is a basic information requirement (entity based only), uses further characteristics in the applicability functionality of the original IDS, or requires an IDS extension. In addition to the information given here, the information requirements specify data types and allowed values for the properties. These can be found in the corresponding IDS files (published in Loibl et al. (2024)).

Different spaces are distinguished using the PredefinedType and the SpaceCategory and StaircaseType properties. In BRISE-Vienna, the property SpaceCategory has been introduced to distinguish between general space categories such as flats, operational units, connection rooms and traffic areas. It already allows the unambiguous identification of flats, but not of safe staircases and parking spots. Safe staircases belong to the category of "Connection room". So do corridors or staircases without high fire resistance requirements. So we need an additional property to define the type of staircase. However, this property is only required for spaces of the category "Connection room", which can be defined in the applicability of an IDS. Parking spots belong to the "Traffic Area" category, as does the space of the entire parking garage. In BRISE-Vienna it was decided to use the existing predefined type PARKING to identify parking spots. All other spaces require the predefined type NOTDEFINED.

In order to regenerate a selected escape route network in the escape route validation concept, doors need properties that relate to the next door on the escape route. In BRISE-Vienna we have introduced the IDDoor and IDNextDoor properties to represent this relationship. However, it should be noted that only doors on escape routes require these properties for escape route validation (second automated step of the escape route analysis). The relevant doors cannot be formally defined in the information requirements, as these are defined after the escape route generation (first automated step of the escape route analysis). Therefore, as a logical check, we have specified that at least one door in the model must have these properties. This is specified in IDS by the cardinality (necessity or prohibition of elements or characteristics).

Finally, as escape routes start at the flat entrance doors, information about the number of people staying in the flat needs to be provided. This is covered by the new OccupancyNumberFlat property. However, the challenge remains in identifying the flat entrance doors, as only these doors require the property. A flat entrance door has on one side a space of category flat and on the other side a space of any other category. This requires finding the spaces adjacent to the door and then comparing their space category (see Fig. 1). IFC provides the relation IfcRelSpaceBoundary to identify the elements that bound a space, such as walls, slabs or doors. It can be used in reverse to get the spaces adjacent to a door. While the previous information requirements are covered by the IDS functionality, this one requires an extension to IDS. IDS includes the PartOf facet to cover relations between elements. However, the PartOf facet is currently restricted to six specific relations and does not include IfcRelSpaceBoundary. In addition, the PartOf facet can require a relation and the entity of the element related by that relation. Additional requirements, such as the value of a property of the related element, are not possible. Finally, the relation IfcRelSpaceBoundary finds several elements. These elements must be compared with each other to determine whether a door is a flat entrance door.

Table 1
Information requirements for the escape route analysis of Fischer et al. (2023).

Applicability	Requirements
IfcSpace	IfcSpace.PredefinedType
IfcSpace	ZDB_SpaceData.SpaceCategory
IfcSpace SpaceCategory = "Connection room"	ZDB_SpaceData.StaircaseType
IfcSpace PredefinedType = "NOTDEFINED" SpaceCategory = "Operational unit, Connection room, Traffic area"	Pset_SpaceOccupancyRequirements.OccupancyNumberPeak
IfcDoor Cardinality: is defined for at least one	ZDB_DoorData.IDDoor
IfcDoor Cardinality: is defined for at least one	ZDB_DoorData.IDNextDoor
IfcDoor SpaceCategory of one bounding space = "Flat" and for another ≠ "Flat"	ZDB_DoorData.OccupancyNumberFlat

classic information requirements (entity-based)
 IDS applicability used
 IDS extension required

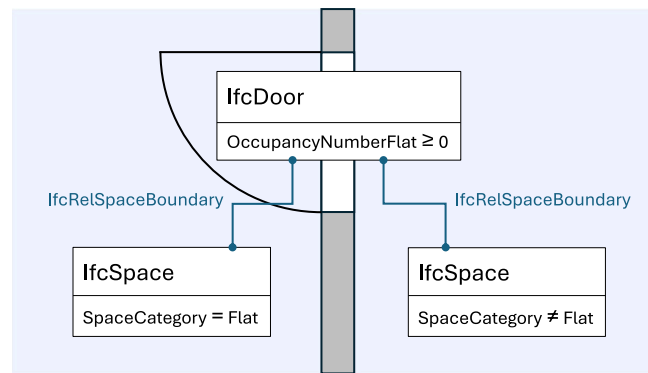


Fig. 1. Identification and requirements of flat entrance doors depending on the category of the related spaces.

2.2. Analysis of regulations requiring similar functionalities

The identified requirements for IDS to define information of elements dependent on related elements have the potential to implement simple code compliance checks. Simple code compliance checks are considered here to be checks on information that is explicitly defined in the IFC file, such as attributes, properties or relations. The facets of the IDS provide the general functionality to process this information. The required specific functionality of each facet depends on the characteristics of the different regulations. Conversely, there are checks that necessitate complex computation based on explicit information, such as in the case of escape route analysis (Fischer et al., 2023). However, this is clearly beyond the scope of IDS and therefore beyond the scope of this work. In this research, we demonstrate the extension of IDS for code compliance checks with Austrian regulations, which require similar functionality to that required for the determination of flat entrance doors for escape route analysis. The findings of the considered regulations are subsequently employed to examine the general potential of IDS for code compliance checking. Suitable regulations can be found, for example, in the OIB 2 guideline (Austrian Institute of Construction Engineering, 2023a) for the related topic of fire resistance of elements. Table 1b of OIB 2 defines general fire resistance requirements for elements, distinguishing for example between load-bearing elements in general (walls, slabs, columns and beams), partition walls and slabs between different units, and fire compartment walls and slabs. Table 2a of OIB 2 concerns the fire resistance of elements in safe staircases along escape routes. The original version of IDS allows the verification

of simple requirements, such as the definition of possible fire resistance values. However, OIB 2 defines these requirements in relation to several other characteristics of the building. Table 2 shows an excerpt from Table 1b of OIB 2 on wall and slab requirements, which includes relevant dependencies and challenges.

Two challenges for the representation in IDS are related to the assignment of building elements such as walls and slabs to the spatial structure of the building. The spatial structure is hierarchical and contains elements from IfcSite, IfcBuilding and IfcBuildingStorey. Building elements such as walls and slabs are usually assigned to the spatial structure at the building storey level. The first challenge concerns the position of the building element. The requirements distinguish between elements on the top storey, other above-ground storeys and underground storeys. Whether an element is above or below ground is not explicitly specified for the building elements. However, the IFC standard provides an "AboveGround" property in the Pset_BuildingStoreyCommon property set for building storeys. The top storey can also be identified by a property. In the BRISE-Vienna research project, an "EscapeLevel" property was introduced to specify the escape height of the top storey. This property is only required for the top storey and can therefore be used to distinguish it from other above-ground storeys. It is consequently necessary to identify the associated storey from the wall or slab and check its properties. The relation that defines the containment of building elements in the spatial structure is IfcRelContainedInSpatialStructure, which is already covered by the relations of the PartOf facet of IDS. Then, requirements must be defined for the property of the retrieved building storey. The second challenge

Table 2

Excerpt of Table 1b of the OIB 2 guideline showing partition wall and slab requirements.

Building class (in Austria: Gebäudeklasse - GK)	GK 2	GK 4
2 Partition walls between different units (excluding safe staircase walls)		
2.1 in the top storey	REI 30 or EI 30	REI 60 or EI 60
2.2 in other above-ground storeys	REI 30 or EI 30	REI 60 or EI 60
2.3 in underground storeys	REI 60 or EI 60	REI 90 and A2 or EI 90 and A2
4 Slabs and sloping roofs with a slope $\leq 60^\circ$		
4.1 Slabs above the top storey	R 30	R 30
4.2 Partition slabs above the top storey	REI 30	REI 60
4.3 Partition slabs above above-ground storeys	REI 30	REI 60
4.4 Slabs inside flats or operational units in above-ground storeys	R 30	R 30
4.5 Slabs above underground storeys	REI 60	REI 90 and A2

is the building class (in Austria: Gebäudeklasse — GK) dependency. The building class is defined in the OIB guidelines, describing a characteristic of a building that represents its size and complexity. For example, a detached house corresponds to building class 1, while a residential building with an escape height of up to 11 m corresponds to building class 4. It is therefore appropriate to store this information in a property of the `IfcBuilding` entity. Unlike the relationship between a building element and the building storey, `IfcBuilding` is only indirectly related to the building element. To establish a dependency on the `IfcBuilding` for a requirement concerning the `IfcWall` or `IfcSlab`, it is necessary to use the relation `IfcRelContainedInSpatialStructure` to retrieve the `IfcBuildingStorey` and then use the relation `IfcRelAggregates` (also already covered by IDS) to iterate through the spatial structure and retrieve the `IfcBuilding`. This functionality is already covered by the `PartOf` facet. The `PartOf` facet applies the specified relation recursively until an element of the specified entity is found. If no relation is specified (optional parameter), the `PartOf` facet recursively applies all available relations to identify an element of the specified entity. This covers the required combination of `IfcRelContainedInSpatialStructure` and `IfcRelAggregates` to retrieve the `IfcBuilding` from the `IfcWall` or `IfcSlab`.

A further challenge is to distinguish between partition elements and elements within a unit (such as a flat or an operational unit). In general, this challenge is identical to the one identified for the information requirements of the flat entrance doors for the escape route analysis. It is necessary to retrieve the spaces adjacent to the wall or slab and compare their unit numbers. The unit numbers do not need to match any particular value. Instead, it is only necessary that all unit numbers are the same (for slabs within a unit) or at least one unit number is different (for partition elements). Fig. 2 summarises the distinction using slabs as an example.

Finally, regulation 2 in Table 2 explicitly excludes walls of safe staircases, even though they are also partition walls between different units. It is therefore necessary to ensure that none of the associated elements (spaces) are part of a safe staircase.

Table 2a of OIB 2 for the fire resistance of elements in safe staircases contains similar required functionalities as Table 1b. All defined requirements depend on the building class. In addition, the fire resistance requirements for elements in underground storeys are higher than those for elements in above-ground storeys. However, the requirements for doors also depend on the space category they connect to the staircase. Thus, the space category for the spaces on either side of the door is defined. This is different from the previous definitions, which also required a comparison of the characteristics of related spaces. For the determination of partition elements in Table 1b of OIB 2, it was only relevant if the values were equal or different, the actual values were irrelevant. When determining flat entrance doors for escape route analysis, only the value for the space on one side of the doors was defined, the other side just had to be different. This requirement for

doors of safe staircases in Table 2a of OIB 2 requires not only that the difference in property values be compared, but also that the spaces on either side of the door meet certain property values.

2.3. Formal definition of IDS extensions

The analysis of two additional IDS use cases, namely the definition of information requirements for escape route analysis and the code compliance checking of fire resistance requirements of the Austrian OIB 2 guideline, revealed that elements must be filtered more specifically than is currently possible with IDS. Three potential IDS extensions were identified to cover that functionality:

- Restrictions for related elements,
- consideration of a relation between elements and adjoining spaces, and
- processing and comparing multiple related elements.

These extensions are linked to existing features of IDS and are repeated for several statements, thus promising great potential for extending IDS. The extensions build on each other. The core extension and basis for the others is the capacity to define restrictions based on attributes and properties of related elements in the applicability of IDS. Related elements are linked to the element in question by an IFC relation element (`IfcRel...`). Examples are adjacent spaces or the related building storey. Currently, it is not possible in IDS to define restrictions for these elements, e.g. to define required attributes or properties. The introduction of this functionality necessitates the adaptation of the `PartOf` facet, which deals with relations. Integrating the `Attribute` and `Property` facets within the `PartOf` facet covers the required functionality (see Fig. 3). This enables the definition of attribute and property requirements for the related elements, such as the related building storey must have a property “AboveGround”. Furthermore, as previously stated, the `PartOf` facet has already implemented recursion for the covered relations. Consequently, the extension also allows the definition of requirements for indirectly related elements (e.g. `IfcBuilding` is indirectly related to `IfcWall` through `IfcBuildingStorey`), irrespective of the degree of the relation.

The second extension concerns the current restriction of the `PartOf` facet to six different relations. While `IfcRelContainedInSpatialStructure` and `IfcRelAggregates`, which are required to retrieve building storeys and the building, are included, the relation between spaces and their adjacent elements is not yet considered: `IfcRelSpaceBoundary`. `IfcRelSpaceBoundary` is needed to find the spaces next to doors, walls or slabs to check if these elements divide different units or flats (see Fig. 4). The introduction of `IfcRelSpaceBoundary` as the seventh relation in the `PartOf` facet provides the required functionality. However, this only works if the desired element directly bounds the spaces on both sides of the element. Problems arise for multi-layered elements where each layer is modelled as a separate element. This is the case

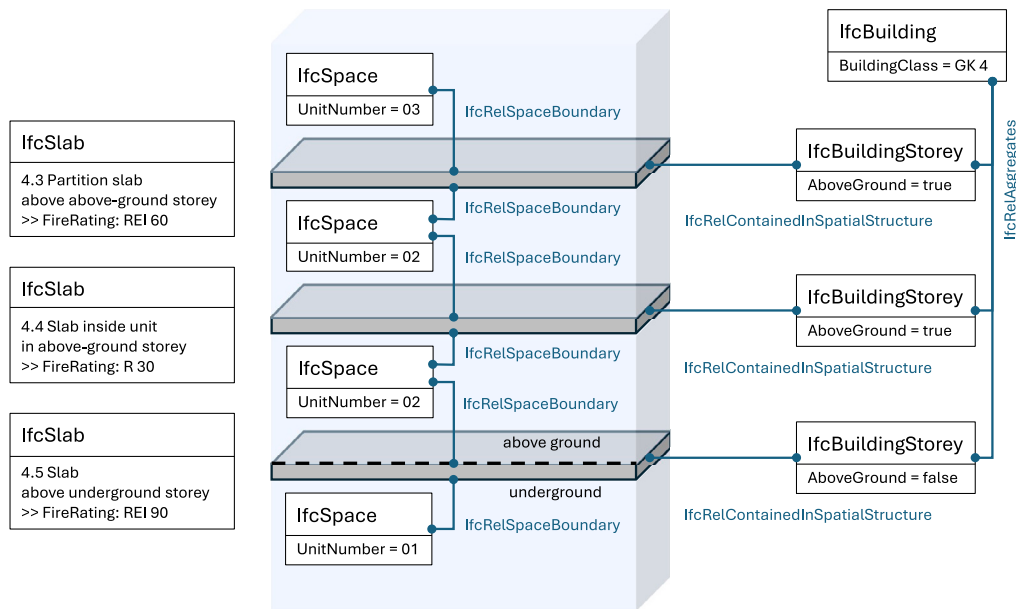


Fig. 2. Identification and requirements of slabs depending on characteristics of related elements.

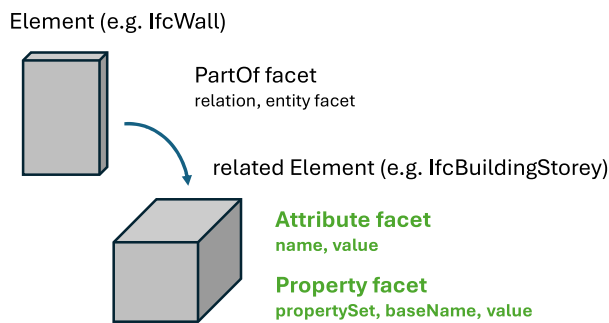


Fig. 3. Concept of including the Property facet and Attribute facet (green) into the PartOf facet.

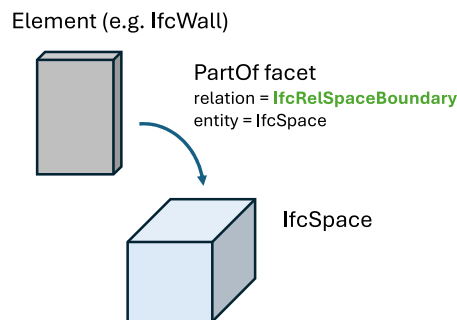


Fig. 4. Concept of including the relation IfcRelSpaceBoundary (green) into the PartOf facet.

with slabs. While the load-bearing slab is modelled as an IfcSlab, the floorings and ceilings are modelled as individual IfcCoverings. None of the three individual elements has a relation to spaces on both sides. Doors and walls do not cause this problem. While doors are single elements, it is also common for walls to be modelled as multi-layered elements (in Austria this is explicitly defined in the ÖNORM A 6241-2 standard (Austrian Standards International, 2015)).

However, the combination of the IfcRelSpaceBoundary relation and the restrictions on related elements is not sufficient to determine whether a door, wall or slab separates two units. In addition, the properties of the related spaces need to be checked together or compared, rather than checked individually (see Fig. 5). The previous requirements analysis revealed four different options for this check or comparison. Either the values must be the same for each related space (for slabs within a unit), at least one value is different (for partition walls or slabs between different units), two different defined values must be present (for doors between staircases on one side and corridors, flats or operational units on the other side), or none of the related spaces may meet the property value (exclusion of walls of safe staircases). Therefore, it is necessary to introduce a new IDS element that defines how the comparison between several retrieved adjacent elements should be performed.

2.4. Adaptation of the IDS schema

The inclusion of the identified functionalities in IDS requires the adaptation of the IDS XML Schema (XSD). The three identified extensions all relate to the functionality of the PartOf facet. This is defined in the schema definition by the complex type partOfType. In its original state, partOfType contains the Entity facet as an XML element (mandatory) to define the IFC entity of the related element and the desired relation as an XML attribute (optional).

The contained relations are defined by the simple type “relations”, which contains the names of the relations as strings. To consider a new relation, the name of the relation can be added to the enumeration, as we did with IfcRelSpaceBoundary (see Fig. 6). The purpose of this extension is to retrieve spaces adjacent to partition elements. buildingSMART International describes space boundaries as physical boundaries of spaces in IFC models (buildingSMART International, 2024a). This refers to a variety of entities such as IfcWall, IfcSlab and IfcDoor. However, requiring the existence of a space next to partition elements is trivial as most IFC models have spaces on at least one side of these partition elements. A stronger functionality is to check and compare the properties of the related spaces. This allows the purpose of the partition element to be identified.

The adaption of the complex type partOfType of the IDS schema definition enables the functionality of checking properties of related elements. In addition to the Entity facet, the Attribute and Property

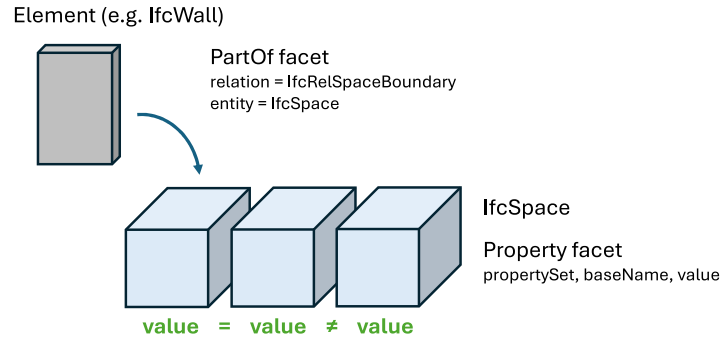


Fig. 5. Concept of checking and comparing multiple related elements (green).

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<xs:simpleType name="relations">
  <xs:restriction base="xs:string">
    <xs:enumeration value="IFCRELAGGREGATES"/>
    <xs:enumeration value="IFCRELASSIGNSTOGROUP"/>
    <xs:enumeration value="IFCRELCONTAINEDINSPATIALSTRUCTURE"/>
    <xs:enumeration value="IFCRELNESTS"/>
    <xs:enumeration value="IFCRELVOIDSELEMENT IFCRELFILLSELEMENT"/>
    <!-- Extension from the authors -->
    <xs:enumeration value="IFCRELSPACEBOUNDARY"/>
  </xs:restriction>
</xs:simpleType>

```

Fig. 6. Inclusion of IfcRelSpaceBoundary into the IDS schema definition.

```

<xs:complexType name="partOfType">
  <xs:sequence>
    <xs:element name="entity" type="ids:entityType" minOccurs="1"/>
    <!-- Extension from the authors -->
    <xs:element name="attribute" type="ids:attributeType"
      minOccurs="0" maxOccurs="unbounded"/>
    <!-- Extension from the authors -->
    <xs:element name="property" type="ids:propertyType"
      minOccurs="0" maxOccurs="unbounded"/>
    <!-- Extension from the authors -->
    <xs:element name="multiElementProcessing" type="ids:multiElementProcessingType"
      minOccurs="0" maxOccurs="1"/>
  </xs:sequence>
  <xs:attribute name="relation" type="ids:relations" use="optional"/>
</xs:complexType>

```

Fig. 7. Adaption of the PartOf facet in the IDS schema definition.

facets are also integrated into the PartOf facet (see Fig. 7). Unlike the Entity facet, the two new facets are set optional (minOccurs=0). This ensures that the PartOf facet can be used as originally defined. In addition, the number of Attribute and Property facets is not limited, allowing the specification of multiple characteristics to be satisfied by the related elements.

Finally, in addition to checking the characteristics of related elements, the functionality to compare the characteristics of different related elements is to be implemented. We introduce a new simple XML element called multiElementProcessing for this purpose. The idea of comparing characteristics of related elements is applicable to both attributes and properties. However, our use cases only included restrictions for properties. Therefore, we introduced the functionality only for properties and added the new element in the sequence after the Property facet, as it is related to it.

The value of the element specifies how the multiple elements retrieved are to be checked and compared. The possible values are defined by the newly introduced simple type multiElementProcessingType (see Fig. 8). Corresponding to the possibilities of how to check and compare the elements identified in the previous analysis, five different values are valid to use: allElementsMustComply, noElementMayComply, samePropertyValuesForAllElements, differentPropertyValuesForAnyElement and differentElementsForEachProperty. So there are 6 different scenarios for checking and comparing multiple

elements: using one of the five keywords or not using the new element. For the six scenarios, the number of property facets involved and the possibility or necessity to define the property value are different. Table 3 lists the scenarios and the corresponding Property facet and property value options.

In the first case, the new element multiElementProcessing is not used. In this case, the standard IDS property check is applied to each element individually. In addition, each specified property facet is processed individually. Thus, at least one of the related elements must satisfy the requirements of all the Property facets contained in the PartOf facet.

The allElementsMustComply and noElementMayComply keywords extend the standard property check by considering all related elements as a group. The two keywords handle two opposite extreme scenarios: all elements must satisfy the property facets (allElementsMustComply) and none of the elements may satisfy the property facets (noElementMayComply). When multiple property facets are given, the conditions must be met for each of them. Thus, these keywords do not compare the values of different components, but check them as a group. The noElementMayComply keyword is required to exclude elements due to the properties of related elements, such as walls of safe staircases from partition walls between different units (Table 1b, regulation 2.2). The keyword allElementsMustComply is not used for the use cases considered, but has been included as a counterpart to noElementMayComply.

```

<!-- Extension from the authors -->
<xs:simpleType name="multiElementProcessingType">
  <xs:restriction base="xs:string">
    <xs:enumeration value="allElementsMustComply"/>
    <xs:enumeration value="noElementMayComply"/>
    <xs:enumeration value="samePropertyValuesForAllElements"/>
    <xs:enumeration value="differentPropertyValuesForAnyElement"/>
    <xs:enumeration value="differentElementsForEachProperty"/>
  </xs:restriction>
</xs:simpleType>

```

Fig. 8. Newly defined type multiElementProcessingType in the IDS schema definition including the allowed values.

Table 3

Overview of the multiple possible scenarios of multiElementProcessing.

Keyword	Property facets	Specify property values
no use	multiple possible	possible
allElementsMustComply	multiple possible	possible
noElementMayComply	multiple possible	possible
samePropertyValuesForAllElements	multiple possible	not possible
differentPropertyValuesForAnyElement	multiple possible	possible
differentElementsForEachPropertyValue	multiple required	required

The samePropertyValuesForAllElements keyword indicates that the property values of different related elements are compared for equality. The important information is whether the values of a particular property are the same for the different related elements. Therefore, the specification of a property value is not intended, as the actual value is not important. The number of Property facets included is not limited, as long as at least one is defined. If multiple Property facets are defined, all related elements must provide the same values for all specified properties. The functionality of samePropertyValuesForAllElements is required for the identification of slabs within a flat or operational unit as defined in regulation 4.4 of Table 1b.

The differentPropertyValuesForAnyElement keyword is the opposite of samePropertyValuesForAllElements and compares the property values for differences. If the value of at least one related element is different from the others, the condition of this keyword is fulfilled. Again, it is the difference between the values that counts, not the actual value. However, it is also possible to define a specific value. In this case, the specified value must be met by at least one related element and not by at least one other related element to ensure the difference. This functionality is necessary to identify flat entrance doors or walls of safe staircases. The property value on one side of the partition element is defined (e.g. flat or safe staircase) but is arbitrary on the other side. As in the previous cases, it is possible to use multiple Property facets. The condition must then be satisfied for each Property facet individually.

The last case is the use of the differentElementsForEachProperty keyword. This case requires the specification of at least two Property facets with specified property values. The aim is to find an element that satisfies the requirements of the first Property facet and another for the second. In effect, the matching of the defined values is checked, but by different elements. If more than two Property facets are specified, different elements must be found for each of them. The differentElementsForEachProperty keyword with the definition of two Property facets allows to identify partition elements with a certain space on one side and a certain space on the other side, like doors between safe staircases and flats.

2.5. Prototype software implementation

The change to the IDS schema means that existing software is unable to process IDS files including the extensions described. Therefore, we have implemented a prototype that covers the required functionality as a proof of concept. The prototype was developed using the open source IfcOpenShell library (version 0.7.0) (IfcOpenShell, 2024) for the Python programming language. IfcOpenShell includes an additional package called IfcTester for processing IDS files. IfcTester fully covers the functionality of the six IDS facets and allows IDS files to be created and IFC

files to be validated against IDS files. In addition, IfcTester includes reporting functionality that is linked to the BCF (BIM Collaboration Format [buildingSMART International, 2024c](#)) package of IfcOpenShell, allowing the results to be stored as check reports in BCF and viewed in existing software.

In order to implement an IDS checker with the required functionality, three core adaptations to the IfcTester were necessary: processing the Attribute and Property facets within the PartOf facet, including IfcRelSpaceBoundary as a new relation, and distinguishing between the cases of comparing the related elements of IfcRelSpaceBoundary. For all three adaptations, existing code was reused, such as the existing functions for checking the Attribute and Property facets. The results were exported to BCF files (version 2.1). Two BCF files were created for each specification (published in [Loibl et al. \(2024\)](#)). The first specifies the elements that failed the check, i.e. the requirements defined in the IDS file are not met. This file is a classic checking report. The second file lists the applicable elements and is used to validate the applicability of the IDS specifications and visualise the results of our extension. Since the presented IDS extensions all concern the applicability of an IDS, this BCF file is crucial for the validation of our approach.

2.6. Validation of the concept

The proposed IDS extensions were validated by checking two BIM models from the BRISE-Vienna project against the generated IDS files. Both models are publicly available. One is a custom test model ([Fischer et al., 2024c](#)) created to validate the escape route analysis of [Fischer et al. \(2023\)](#) (see Fig. 9 left). Initially, the model only contained information relevant to the escape route analysis. For this study, we extended it with information relevant to the OIB 2 tables considered and we included test cases for each specification. The second model is a real-world Building Application Model ([Fischer et al., 2024b](#)) provided during the pilot phase of BRISE-Vienna (see Fig. 9 right). For this study, we removed all non-standard information that was not necessary for our use case and translated the German terms into English.

We verified the functionality of our concept by importing the generated BCF files into the BIM checking software Solibri Office ([Solibri Inc, 2024](#)). This made it easy to visually check whether the expected applicable elements were found.

3. Results

The extension of the IDS schema enabled the creation of specifications for the alphanumeric information requirements for escape route analysis and code compliance checking of the considered fire resistance

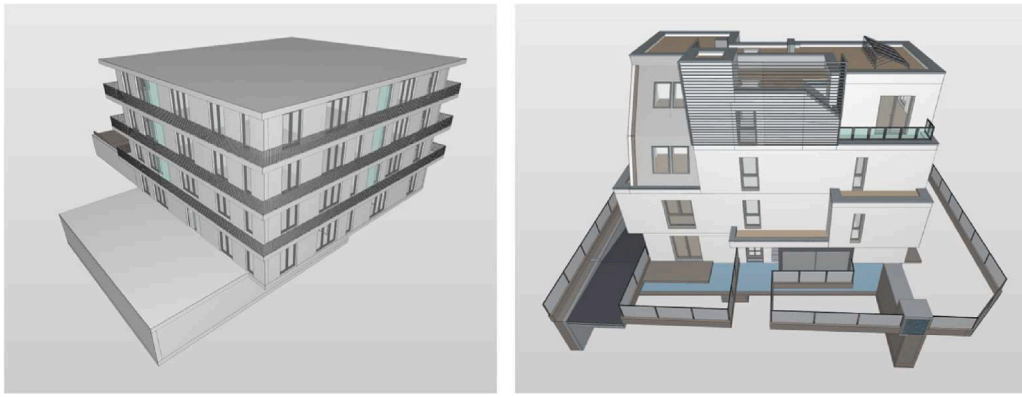


Fig. 9. 3D view of the custom test model (left) and the real-world test model (right).

Table 4
Use of the extensions in the different IDS files.

	Related element restrictions	IfcRelSpaceBoundary	Multi element processing
Escape route Analysis	1	1	1
OIB 2 Table 1b	66	13	12
OIB 2 Table 2a	17	5	3
	84	19	16

regulations of the OIB 2 guideline in the Tables 1b and 2a. These specifications were divided into three IDS files (Loibl et al., 2024), one for the escape route analysis and one each for the OIB 2 tables. The proposed extensions are intended to define the applicable elements of the specifications more specifically. Consequently, the excerpts of the IDS specifications included in this section only show the applicability parts of the specifications. However, the extensions can be used identically for elaborate definitions in the requirement parts of IDS specifications. Table 4 summarises the occurrences of the extensions for the different IDS files. While the information requirements for escape route analysis use each extension once, the code compliance checks for the OIB 2 tables rely on the related element restrictions in particular. This is due to the dependency on the building class of all regulations, which is a property stored in the related IfcBuilding. IfcRelSpaceBoundary and multi-element processing occur for specifications concerning elements between particular spaces, such as partition walls or slabs between different units. In the following section, the IDS specifications and checking results of the two considered use cases will be described in detail.

3.1. Results for the information requirements of the escape route analysis

The information requirements for escape route analysis are defined by five IDS specifications. Specification 1 contains two classic information requirements that are only entity based (lines 1 and 2 of Table 1). All IfcSpace elements require the property SpaceCategory and a predefined type, both with values from defined lists. Specifications 2 to 4 cover requirements using additional IDS facets in the applicability to specify the elements concerned (lines 3 to 6 of Table 1). All IfcSpace with SpaceCategory “Operational unit”, “Connection room”, or “Traffic area” require the property OccupancyNumberPeak because they are possible escape route starting positions. Connection rooms additionally require the property StaircaseType to identify safe staircases. There must also be doors with the properties IDDoor and IDNextDoor to predefine the course of escape routes. These first four specifications are covered by the standard IDS schema and can be checked by existing IDS checkers. Fig. 10 shows a comparison of the check results in the IfcOpenShell IfcTester and usBIM.IDS (ACCA software, 2024). Both find the same number of applicable and failed elements per specification

(IfcTester shows the number of passed elements while usBIM.IDS shows the number of failed elements).

Specification 5 uses the three extensions in its applicability to identify flat entrance doors (line 7 of Table 1). Fig. 11 shows an excerpt of the applicability of the IDS specification. The Entity facet, which specifies the IfcDoor, is followed by a PartOf facet that uses the IfcRelSpaceBoundary relation to retrieve the spaces bounded by the door in question. Then, within the PartOf facet, a Property facet is used to define the space category of the related space. Finally, the new IDS element multiElementProcessing defines how to compare the values of multiple spaces found. The differentPropertyValuesForAnyElement keyword indicates that at least one element must have a different value than the others. The combination with the defined property value “Flat” specifies that one of the spaces must be of the category flat. This ensures that the IfcDoor bounds at least one space of category Flat and one space of another category, and is therefore a flat entrance door.

We tested this specification using the extended IDS checker developed based on IfcTester from IfcOpenShell. For both BIM models used, the flat entrance doors were correctly detected. Fig. 12 shows the applicable elements (green) referenced in the BCF files imported into Solibri Office. The custom test model has 4 flats on each of the top two floors (= 8 flat entrance doors). The real-world model contains 2 flats on each of floors 0 to 2 and one on the top floor (= 7 flat entrance doors). The results demonstrate the feasibility of identifying flat entrance doors for escape route analysis by defining an extended IDS applicability using the proposed extensions. This approach allows the information requirement for the flat entrance doors (OccupancyNumberFlat property) to be specifically defined.

3.2. Code compliance checking of tables of the OIB 2

The presented IDS extensions allow the representation of many regulations of Tables 1b and 2a of the OIB 2 guidelines. Detailed results are shown for the example of partition walls between different units according to regulation 2.2 of Table 1b. Four PartOf facets are required to identify the relevant partition walls. The first two PartOf facets deal with relations to elements of the spatial structure of the model (see Fig. 13). The first PartOf facet covers the building class dependency. It links the wall to an element of the IfcBuilding entity without declaring the required relation. The IDS tries the implemented relations recursively until the desired entity is found. This makes it possible to first use IfcRelContainedInSpatialStructure to retrieve the building storey and then IfcRelAggregate to retrieve the building within one PartOf facet. The second PartOf facet specifies that the building storey must be above ground.

The third and fourth PartOf facets define dependencies on adjacent spaces (see Fig. 14). The third PartOf facet excludes walls of safe staircases from the applicable elements. The IfcRelSpaceBoundary relation links the walls to spaces. The included Property facet specifies the

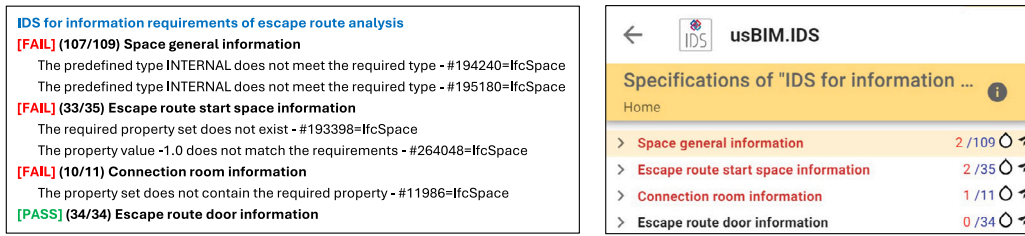


Fig. 10. Checking results for the standard IDS specifications of the escape route analysis using the IfcTester (left) and usBIM.IDS (right).

```
<specification name="Flat entrance door information" ifcVersion="IFC4">
  <applicability minOccurs="0" maxOccurs="unbounded">
    <entity>
      <name>
        <simpleValue>IFCD00R</simpleValue>
      </name>
    </entity>
    <partOf relation="IFCRELSPACEBOUNDARY">
      <entity>
        <name>
          <simpleValue>IFCSPACE</simpleValue>
        </name>
      </entity>
      <property dataType="IFCLABEL">
        <propertySet>
          <simpleValue>ZDB_SpaceData</simpleValue>
        </propertySet>
        <baseName>
          <simpleValue>SpaceCategory</simpleValue>
        </baseName>
        <value>
          <simpleValue>Flat</simpleValue>
        </value>
      </property>
      <multiElementProcessing>differentPropertyValuesForAnyElement</multiElementProcessing>
    </partOf>
  </applicability>
</specification>
```

Fig. 11. Definition of flat entrance doors in the IDS applicability.

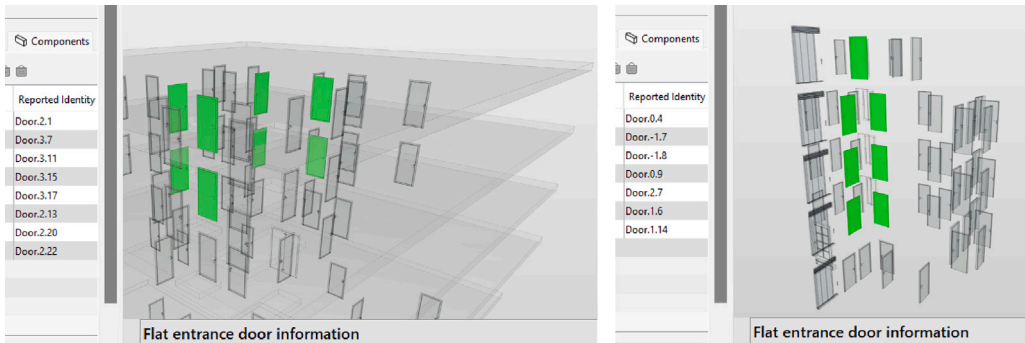


Fig. 12. Identification of flat entrance doors (green) in Solibri Office for the custom test model (left) and the real-world model (right).

relevant property to identify safe staircases. Finally, the `noElementMayComply` keyword of `multiElementProcessing` allows only elements that do not comply with the defined Property facet. The fourth `PartOf` facet specifies how partition walls between different units are to be detected. Again, the relation `IfcRelSpaceBoundary` is used to retrieve adjacent spaces. The combination of a Property facet and the new IDS element `multiElementProcessing` then defines that the property value of "UnitNumber" must be different for at least one of the related spaces. The keyword for this is `differentPropertyValuesForAnyElement`. The definition only targets a difference between the values. The actual values of the properties are irrelevant.

Fig. 15 shows a top view of the expected result (left) and the identified applicable walls (right) of the custom test model. On each of the top three floors there are four flats or operational units and a central staircase. The identified partition walls clearly show the different units. As defined in regulation 2.2 of Table 1b, walls of safe staircases are

excluded. However, the result does not fully represent regulation 2.2 of Table 1b. This rule applies to all storeys above ground except the top storey. While regulation 2.1 (for the top storey) can be implemented by additionally requiring the building storey to have the property "EscapeLevel", it is not possible to include only storeys that do not have the property (for regulation 2.2). One potential solution is to require the "EscapeLevel" property for all other storeys as well, with values such as "not defined" assigned to them. This would enable the identification of other storeys above ground, other than the top storey. However, the objective is to require properties only for elements that genuinely require them, rather than assigning properties to all elements within an entity. Another solution is to introduce exclusion or prohibition of elements in the applicability. Exclusion or prohibition is covered in IDS by cardinality, which only exists for whole specifications or for facets used in requirements. The introduction of cardinality in the applicability of an IDS, identical to its functionality in the requirements, would allow

```

<specification name="2.2 Partition walls on above-ground storeys (excluding staircase walls)"
  ifcVersion="IFC4">
  <applicability>
    <entity>
      <name>
        <simpleValue>IFCWALL</simpleValue>
      </name>
    </entity>
    <partOf>
      <entity>
        <name>
          <simpleValue>IFCBUILDING</simpleValue>
        </name>
      </entity>
      <property dataType="IFCLABEL">
        <propertySet>
          <simpleValue>ZDB_BuildingData</simpleValue>
        </propertySet>
        <baseName>
          <simpleValue>BuildingClass</simpleValue>
        </baseName>
        <value>
          <simpleValue>GK4</simpleValue>
        </value>
      </property>
    </partOf>
    <partOf>
      <entity>
        <name>
          <simpleValue>IFCBUILDINGSTOREY</simpleValue>
        </name>
      </entity>
      <property dataType="IFCLOGICAL">
        <propertySet>
          <simpleValue>Pset_BuildingStoreyCommon</simpleValue>
        </propertySet>
        <baseName>
          <simpleValue>AboveGround</simpleValue>
        </baseName>
        <value>
          <simpleValue>TRUE</simpleValue>
        </value>
      </property>
    </partOf>
  </applicability>

```

Fig. 13. Dependencies on characteristics of the IfcBuilding and IfcBuildingStorey defined in the IDS applicability.

```

<partOf relation="IFCRELSPACEBOUNDARY">
  <entity>
    <name>
      <simpleValue>IFCSPACE</simpleValue>
    </name>
  </entity>
  <property dataType="IFCLABEL">
    <propertySet>
      <simpleValue>ZDB_SpaceData</simpleValue>
    </propertySet>
    <baseName>
      <simpleValue>StaircaseType</simpleValue>
    </baseName>
    <value>
      <simpleValue>0IB2Tab2ab</simpleValue>
    </value>
  </property>
  <multiElementProcessing>noElementMayComply</multiElementProcessing>
</partOf>
<partOf relation="IFCRELSPACEBOUNDARY">
  <entity>
    <name>
      <simpleValue>IFCSPACE</simpleValue>
    </name>
  </entity>
  <property dataType="IFCLABEL">
    <propertySet>
      <simpleValue>ZDB_SpaceData</simpleValue>
    </propertySet>
    <baseName>
      <simpleValue>UnitNumber</simpleValue>
    </baseName>
  </property>
  <multiElementProcessing>differentPropertyValuesForAnyElement</multiElementProcessing>
</partOf>
</applicability>

```

Fig. 14. IDS applicability to identify partition walls between different units excluding walls of safe staircases by comparing characteristics of adjoining spaces.

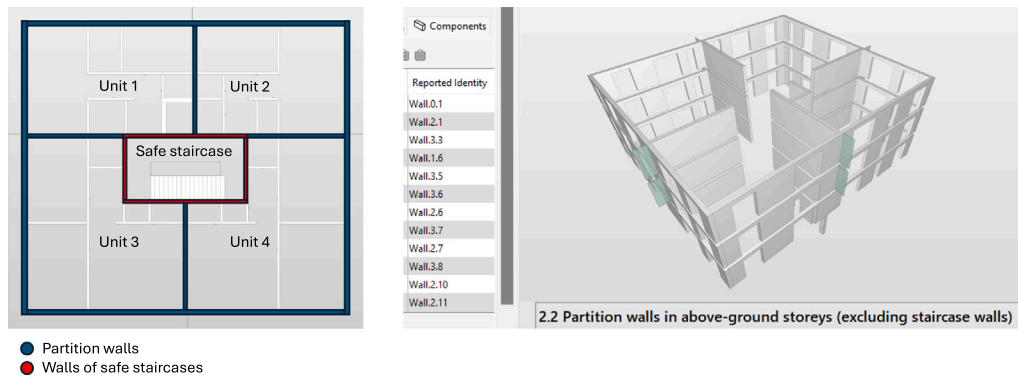


Fig. 15. Identification of partition walls between different units excluding walls of safe staircases for the custom test model in Solibri Office: top view of the expected result (left) and 3D view of the identified walls (right).

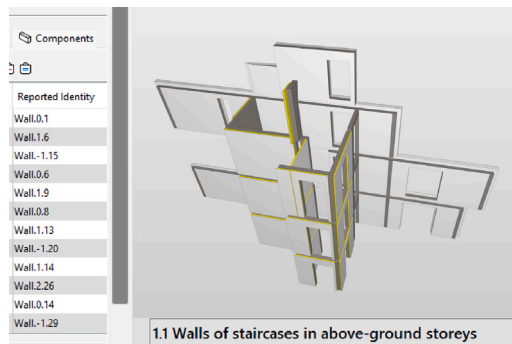


Fig. 16. Incomplete identification of walls of safe staircases for the real-world model in Solibri Office.

the exclusion of building storeys that have the “EscapeLevel” property and thus fully represent the described regulation.

While the code compliance checks using the custom test model produced the expected results, the real-world model revealed a challenge: the quality of the model. An example is shown in Fig. 16 for the results of walls of safe staircases (Table 2a, regulation 1.1). While the walls of safe staircases were sufficiently found in floors 0 to 2, only two walls were found in the top floor. The missing elements are due to missing or uncategorised space boundaries. The reason for the incorrect space boundaries can be attributed to two factors related to modelling errors. Firstly, some multi-layered walls were modelled as individual elements per layer. This means that no single wall has space boundaries on both sides. This problem has been discussed theoretically, but should not occur for walls as the Austrian standard ÖNORM A 6241-2 (Austrian Standards International, 2015) explicitly requires walls to be modelled as multilayer elements. Secondly, the space boundaries were not created by the authoring software or were not related to the correct elements. This is due to inaccurate modelling of the space elements. As authoring software such as Archicad (used for the two models) uses the modelled geometry to derive the space boundaries, accurate modelling is critical to achieving a correct result. In the case of the real-world model, there is a small deviation between the surface of the wall and the space and therefore the software does not generate the correct space boundary. This error occurred in most of the specifications using the IfcRelSpaceBoundary relation. This does not mean that the proposed IDS extension is not applicable, but it does show one of its limitations.

In general, the results showed the potential of the IDS extensions for code compliance checking. However, the extensions could not cover all the regulations defined in the considered tables. Table 5 shows an evaluation of the possibility to cover the building code regulations by the proposed IDS extension.

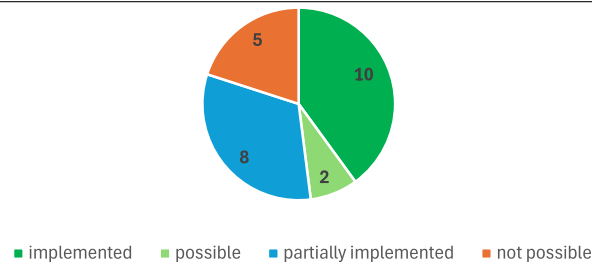
The regulations have been divided into four categories. “Implemented” means that we have created an IDS specification for the regulation and the validation using the developed prototype and the two BIM models was successful. Regulations categorised as “possible” were theoretically analysed but not included in the IDS files due to the lack of relevant elements in the test data. Regulations with limited functionality were categorised as “partially implemented”. This is the case for eight regulations in Table 1b. Regulations 1.2, 2.2 and 4.3 concern elements in above-ground storeys other than the top storey. Due to the use of a property that is only present in the top storey, the top storey can be identified (for regulations 1.1, 2.1, 4.1 and 4.2 in Table 1b), but it cannot be excluded from the other storeys above ground (defining an exception). It is not possible to filter out elements (storeys) that are missing a property. Furthermore, most regulations for slabs (regulations 4 and 5 in Table 1b) require the properties of adjacent spaces to be checked or compared (are they in one unit, do they separate units, or do they border balcony spaces). As discussed in theory, adjacent spaces can only be found if the slabs directly bound the spaces. Therefore, the check only works reliably for slabs without coverings. However, for both BIM models used in this study, the adjacent spaces could still be identified. The custom test model did not include any coverings and therefore did not cause any problems. The real-world model only included floorings, but no ceilings. As the slabs were modelled continuously for the whole storey and each storey contained different units, the slabs bounded spaces of different units (in the same storey). The final category of regulations are those that could not be covered by the proposed IDS extensions. There were two types of barriers. The first are geometric or topological requirements, e.g. whether a wall is on the site boundary or a slab is above a staircase. Positioning information is not covered by relations and is therefore not possible in IDS. The same problem occurs for the determination of stairs or railings in safe staircases (Table 2a, regulations 4 and 5). There is no relation between these elements and the spaces in which they are located. Their containment connects the storey of the building and they are not considered as space boundary elements (at least in the test models). The second obstacle is regulations that are not fully computer interpretable, as is the case for smoke extraction systems.

A total of 12 regulations can be fully covered by the proposed IDS extension. Eight other regulations have limitations, but have been partially implemented. However, five regulations are still not possible with the IDS extension, mostly due to geometrical or topological characteristics. It should be noted that we have concentrated on the requirements concerning the fire resistance of elements. The tables also define requirements for the spread of flames over the surface, which can be implemented identically by using another property (SurfaceSpreadOfFlame).

Table 5

Evaluation of the possibility to define code compliance checks in the extended IDS schema for the OIB 2 Tables 1b and 2a.

Table 1b	
1 load-bearing elements (excluding slabs and fire compartment walls)	
1.1 in the top storey	implemented
1.2 in other above-ground storeys	partially implemented
1.3 in underground storeys	implemented
2 Partition walls between different units (excluding walls of stair cases)	
2.1 in the top storey	implemented
2.2 in above-ground storeys	partially implemented
2.3 in underground storeys	implemented
2.4 between units of terraced houses	possible
3 Fire compartment walls and slabs	
3.1 Fire compartment walls on the site boundary	not possible
3.2 other fire compartment walls and slabs	implemented
4 Slabs and sloping roofs with a slope $\leq 60^\circ$	
4.1 Slabs above the top storey	partially implemented
4.2 Partition slabs above the top storey	partially implemented
4.3 Partition slabs above above-ground storeys	partially implemented
4.4 Slabs inside flats or operational units in above-ground storeys	partially implemented
4.5 Slabs above underground storeys	partially implemented
5 Balcony slabs	partially implemented
Table 2a	
1 Walls of staircases	
1.1 in above-ground storeys	implemented
1.2 in underground storeys	implemented
2 Slab above the staircase	not possible
3 Doors in walls of staircases	
3.1 to flats, operational units or other rooms	implemented
3.2 to corridors in above-ground storeys	implemented
3.3 to corridors in underground storeys	implemented
4 Stair flights and landings in staircases	not possible
5 Railing fillings in staircases	not possible
6 Smoke excavation systems	not possible
7 External stairs	possible



4. Discussion

This research focused on analysing whether IDS could be modified to define specifications depending on the characteristics of related elements and the potential of IDS for code compliance checking. The introduction of related element restrictions was possible by reusing the existing facets (Attribute and Property facets) within the PartOf facet. As we reused existing functionality of the IDS software IfcTester in different places, the adaption of the software was very easy. This suggests that the related element restriction does not hinder the straightforward application and processing of IDS. Furthermore, this extension was the basis for implementing automated code compliance checks for the considered regulations. As all these regulations have dependencies on the characteristics of the building class, the standard IDS schema could not represent any of them. Therefore, the extension was used for each of the implemented checks. The concept of integrating the Attribute and Property facet into another facet also promises potential for the definition of material requirements. The standard Material facet allows specific materials to be required, but does not allow any of their properties to be specified. The availability of material properties is crucial for other use cases such as circular economy. As [Tomczak et al.](#)

(2024) noted in their research on using IDS to capture circularity requirements, IDS is lacking for checking material properties. Integrating the Attribute and Property facet into the Material facet provides this missing functionality.

In addition to the restrictions on related elements, the use cases considered revealed the need for an additional relation in the IDS: IfcRelSpaceBoundary. This was necessary because many requirements were related to adjacent spaces. IfcRelSpaceBoundary showed a strong dependency on the modelling quality of the BIM model. This is because space boundaries represent geometric surfaces that are not explicitly modelled, but derived from the geometry of adjacent elements. If these are imprecise and gaps occur, the space boundaries will be incorrect. A reliable use of IfcRelSpaceBoundary would therefore require a preliminary check that the space boundaries are correctly contained in the model. Another challenge was the modelling of elements consisting of single elements or layers. When these elements are modelled individually, as is common for slabs and their coverings, the individual elements do not have space boundaries on both sides and it is therefore not possible to evaluate whether they are partition elements. A fully reliable determination would require either the introduction of a relation between the structural elements and their coverings in IFC, or the use of geometric methods to check that the individual elements are bonded

together, which is not covered by IDS. The third challenge related to the integration of `IfcRelSpaceBoundary` is the fact that it usually relates multiple elements. Retrieving multiple related spaces is generally necessary, otherwise it would not be possible to determine if a wall or door is between two different space categories. However, it adds the challenge of processing, checking and comparing multiple related elements. This functionality has not yet been considered in IDS. The introduction of the new `multiElementProcessing` element in the `PartOf` facet solved the problem for our use cases, but it also showed that multiple elements can be checked and compared in many different ways. We introduced five keywords that covered the required functionality for our use cases. Each of them had different options on how many `Property` facets could be included and whether a property value should be defined. In fact, each keyword defines a specific behaviour and expected outcome for checking the elements. Each new keyword or term requires an algorithm behind it. This changes the scope of IDS. It turns into a rule language. This step significantly extended the functionality of IDS and enabled sophisticated checks in combination with related spaces, but it made the software implementation of IDS more complicated. However, regarding the fact that also existing relations can retrieve multiple elements, such as `IfcRelAssignsToGroup`, it may be beneficial to consider this functionality in IDS.

The structure of the created IDS files revealed another potential IDS extension. The IDS applicability only allows to specify which elements should be considered. However, in many cases it would be easier or even necessary to specify which elements should not be included (defining an exception). For example, instead of specifying 10 possible values, it would be easier to specify one restricted value. This is particularly important in cases where not all possible values are known, or where not all elements need to have a particular property. This functionality could be covered by introducing cardinality in the applicability part of an IDS in the same way as its functionality in the requirements part. In addition to improving the functionality of the IDS, this inclusion would also streamline the IDS schema. It eliminates the differences in the definition of the applicability type and the requirements type in the IDS schema definition, making the schema simpler and the applicability and requirements of IDS more consistent to use.

In general, the results of our IDS extension showed potential for code compliance checking. With regard to the Requirement, Applicability, Selection, and Exception (RASE) methodology (Noardo et al., 2022a; Macit İlal and Günaydın, 2017; Kincelova et al., 2020) for formal rule representation, the potential IDS extension encompasses all four aspects of formal rule representation. The Requirement, Applicability, and Selection aspects have already been addressed in the standard IDS. The proposed extension, which incorporates the cardinality in the applicability of the IDS, introduces the exception functionality. Our other extensions facilitate a more detailed and targeted definition of the selection and exception. Regulations containing various dependencies like in the considered use cases can be sufficiently checked, as long as the relevant information is explicitly represented by attributes, properties or relations in the IFC. The main extension, the nesting of facets, seems to be beneficial to be applied to other facets besides the `PartOf` facet, as described for the `Material` facet. Extending the nesting functionality to all facets could lead the way to a complete syntax for checking all explicit information contained in the IFC file. By adding all existing relations in the `PartOf` facet, it would be possible to find any related element and define any constraint on it. However, special cases or challenges related to specific relations in IFC, such as those described for `IfcRelSpaceBoundary`, will need to be addressed on a case-by-case basis. Therefore, their integration should always be linked to real use cases, combined with an analysis of the benefits and complexity of the extension. Still outside the scope of IDS as a rule language is information that is not explicitly given and requires prior computation, such as geometric or topological dependencies. This was the case for the identification of elements on the site boundary or the placement of stairs within spaces. This information about the

3D placement of elements is not represented by relations and therefore cannot be processed by IDS. For such functionality, which is currently clearly outside the scope of IDS, an extension to link external services as proposed by Kremer and Beetz (2023) could provide a solution. Another solution is to use `mvdxml`. This format has been introduced by buildingSMART International mainly to define software implementations for IFC (Tomczak et al., 2022) but can also be used for rule checking (Weise et al., 2016). In terms of adoption in the building permit process, the main difference with IDS is that `mvdxml` is less regimented, which allows for more complex functionality, but makes it more difficult to implement.

5. Conclusions

The aim of this research was to extend the functionality of IDS for its implementation in the BIM-based building permit process. In particular, two permit-related use cases of the Vienna Building Authority were considered: the definition of specific information requirements depending on related elements using the example of the checking rule for escape route analysis, and the implementation of simple code compliance checks. We presented and implemented three extensions for IDS and discussed a fourth.

The extensions with the most potential for integration into IDS are the related element restrictions and the definition of cardinality also in the applicability. Both extensions are easy to implement because their functionality is very similar to the scope of the standard IDS. Therefore, they do not negatively affect the simplicity of IDS. Keeping the schema simple is in line with buildingSMART International's design intention for IDS to be very low level and easy to use and implement. In fact, the inclusion of cardinality in the applicability part would actually make the IDS schema simpler, as the structure of the applicability and requirements parts would be harmonised. Furthermore, the ability to exclude certain elements is common in BIM checking software and formal rule representations (Requirement, Applicability, Selection, and Exception (RASE) methodology) (Noardo et al., 2022a; Macit İlal and Günaydın, 2017; Kincelova et al., 2020).

The other two extensions, namely the use of the `IfcRelSpaceBoundary` relation and the processing of multiple related elements, also significantly enhance the functionality of IDS, but are more complicated. The distinction between different possible cases complicates the definition and checking of the requirements. Therefore, further analysis is required before proposing inclusion in the IDS standard. This includes identifying other use cases that depend on related spaces and require the processing of multiple related elements. Furthermore, moving the functionality to a separate extended `PartOf` facet could help to distinguish between standard and extended functionality.

In general, the proposed extensions allowed to define the information requirements for escape route analysis and also showed potential for code compliance checking. In fact, they define a rule language on top of the existing IDS. This introduces a new standard for code compliance checking. Extending the proposed extension from the IDS facets relevant for our use cases to all IDS facets promises potential to create a complete general syntax for checking information explicitly included in the IFC file. However, due to its simplicity, IDS also has clear boundaries for code compliance checking, as clauses that require prior computation are not covered (Nuyts et al., 2024). Other rule-based solutions (Stepien et al., 2023; Ghannad et al., 2019; Weise et al., 2016) provide more functionality for this purpose. However, no approach has yet been established as a standard format for checking rules. In addition, building authorities are already using IDS to define and publish their information requirements. Reusing an already implemented standardised format for another use case helps to keep the building permit process manageable. However, since IDS only provides partial coverage of code compliance checking, a combination with non-standard checking logic is required to handle regulations that require complex computations. Consequently, the aim is to expand the use of

IDS in building authorities to map as much as possible in a standardised way. This enables better comparability and reusability of checking logic between authorities and companies. Future work could include a detailed examination of introducing the nesting functionality for all IDS facets to increase the coverage. Furthermore, additional building codes could be analysed to identify further regulations for which IDS-based compliance checking can be sufficiently applied. In this step, further potential functionalities for IDS will also be analysed.

CRediT authorship contribution statement

Simon Fischer: Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **Harald Urban:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Christian Schranz:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Patrick Loibl:** Visualization, Validation, Software, Methodology, Investigation. **Léon van Berlo:** Writing – review & editing, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data used is publicly available and is cited in the paper.

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